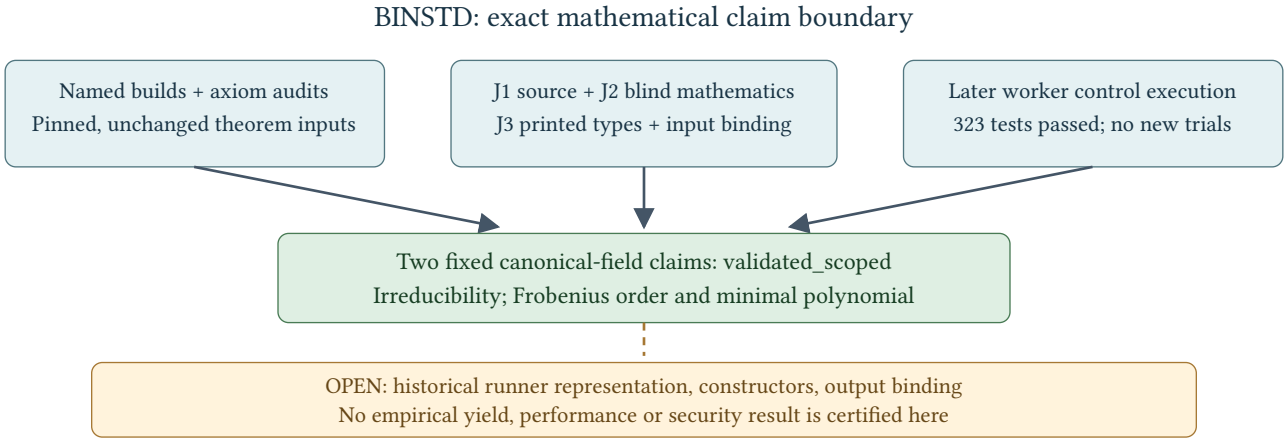
BINSTD elaborated-type supplement

28 October 2026 · Technical review · PR 2629

Scoped outcome: the two fixed mathematical declarations are validated under ordinary Lean/Lake trust. The separately printed types are now independently bound to the frozen claims. No scientific state changes.

Claim	Evidence
Masks 313 and 471 are irreducible over F_2	Named build/audit; J1 source review; J2 blind derivation; J3 printed type and binding
Canonical GF(2^{17}) Frobenius order 17 and stated minimal polynomial	Same three joints; exact canonical field and factor definitions; no open hypotheses



What was checked

Original run 40591043322 passed 89 targets and controls. New run 40636717165 supplies both named type printouts; its overall result remains 88 verified and one nested-toolchain infrastructure failure, with its separate smoke tests skipped after setup failed. J3 matched all four receipt hashes, 65 workspace inputs, ten verifier inputs, the generated 49-module root and ZIP digest. Both named axioms lists contain only propext, Classical.choice and Quot.sound.

A later completed control job, 122925704081 in run 40719320437, reported 323 tests passed using real Lean 4.22. Source comparison binds the unchanged cache worker and session tests; this later evidence does not rewrite the earlier failure or replace the pinned Lean 4.35 mathematics checks.

Independence and limits

The prospective addendum was pushed at dee666eb7d before J3 dispatch. The three-joint checker passes with the previously sealed J2 serialization projection; raw-schema and sparse-checkout failures remain archived. J3's report discloses its lease lapse and runtime metadata limits. No sibling reports were read by J3. No independent kernel rechecker or installed-dependency cleanliness audit was performed.

The runner representation, arithmetic refinement, canonical-field conjugacy, complete invariant-subspace/constructor coverage and archived-output binding remain open. Historical yield and performance graphs are unchanged. Current-head CI and merge gates remain separate delivery obligations.

Sources: inputs.json; reports/TASK-20261028-68ff01/report.md and seal.sha256; machine/repair-kernel-controls.json; machine/independence-composed-materialized.txt. Original round: REVIEW-BINSTD-20261027-b17d. Archive: TASK-20261028-6fe548; Conductor T-753.